

Pawel Renc

Healthcare AI / Foundation Models / ML Systems

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Research profile

Research fellow at Massachusetts General Hospital working on generative models of longitudinal electronic health records and the infrastructure for training and evaluating them. Co-developer of ETHOS, combining healthcare-AI research with industry experience at Snowflake and high-performance computing research at ETH Zürich.

Education

Doctorate · Engineering and Technology	May 2026
AGH University of Krakow · Discipline: Information and Communication Technology	
Dissertation: <i>Foundation model for electronic health records</i> .	
Supervisors: Jarosław Wąs and Arkadiusz Sitek. Degree conferred May 21, 2026.	
Master's degree · Computer Science	2021
AGH University of Science and Technology · System modelling and data analysis	
Bachelor's degree · Computer Science	2020
AGH University of Science and Technology	

Research appointments & experience

Research Fellow	May 2026–present
Massachusetts General Hospital · iPOL	
- Develop EHR foundation models and end-to-end infrastructure for large-scale data processing, training and inference. - Maintain the ETHOS research codebase, spanning tokenization, model training, evaluation and synthetic timeline generation.	
Research Scientist	Oct 2025–May 2026
Snowflake · AI Research	
- Designed and developed an agentic medical-coding system for assigning diagnoses from clinical notes. - Partnered with forward-deployed engineering teams on healthcare applications of medical foundation models.	
AI Research Intern	Apr–Oct 2025
Snowflake	
Developed and fine-tuned EHR foundation models; built training and inference workflows with Transformers, vLLM and Ray.	
Visiting PhD Researcher	Mar 2023–Apr 2025
Massachusetts General Hospital · Harvard Medical School	
- Co-developed ETHOS for generative patient-timeline modeling and zero-shot health prediction; contributed to timeline representation, tokenization, transformer training and inference. - Led software and formal analysis for adaptive risk estimation using ETHOS-generated patient timelines.	
Visiting PhD Researcher	2022–2023
ETH Zürich · Scalable Parallel Computing Laboratory	
Researched GPU execution of graph neural networks and global tensor formulations; coauthored work published at SC 2023.	
PhD Researcher	Jan 2022–Feb 2023
Sano Centre for Computational Medicine	
Developed machine-learning methods for disease-phenotype discovery through patient-data clustering, in collaboration with University of Glasgow clinicians.	

Selected publications

Foundation model of electronic medical records for adaptive risk estimation

First author · *GigaScience*, 2025. DOI: [10.1093/gigascience/giaf107](https://doi.org/10.1093/gigascience/giaf107)

Zero shot health trajectory prediction using transformer

First author · *npj Digital Medicine*, 2024. DOI: [10.1038/s41746-024-01235-0](https://doi.org/10.1038/s41746-024-01235-0)

High-Performance and Programmable Attentional Graph Neural Networks with Global Tensor Formulations

Coauthor · *SC*, 2023. DOI: [10.1145/3581784.3607067](https://doi.org/10.1145/3581784.3607067)

Towards efficient GPGPU Cellular Automata model implementation using persistent active cells

Coauthor · *Journal of Computational Science*, 2022. DOI: [10.1016/j.jocs.2021.101538](https://doi.org/10.1016/j.jocs.2021.101538)

EBIC.JL — an efficient implementation of evolutionary biclustering algorithm in Julia

Coauthor · *GECCO Companion*, 2021. DOI: [10.1145/3449726.3463197](https://doi.org/10.1145/3449726.3463197)

Research software

ETHOS — EHR tokenization, transformer training and health-trajectory inference. Research software supporting generative patient-timeline modeling.

EBIC.jl — GPU-enabled evolutionary biclustering in Julia; developed during master's research.

Selected service & teaching

Peer review: IEEE Journal of Biomedical and Health Informatics (two rounds, 2025); IEEE Transactions on Medical Imaging (2023); BMJ Digital Health & AI (2026).

Grand rounds: Applications of Machine Learning in Healthcare, DHR Health, April 29, 2024.

Research and Teaching Assistant, AGH

2020–2021

Taught Unix, Operations Research and Discrete Systems Modeling.

Technical expertise

Modeling: EHR foundation models, generative AI, transformers, graph neural networks.

ML systems: PyTorch, Transformers, vLLM, Ray; distributed training and inference.

Engineering: Python, C/C++, CUDA, Julia, Bash; Polars, SQL, Linux and GPU computing.

Earlier engineering experience

QED Software · Data Scientist

2020–2021

Built ML methods for detecting erroneous table joins and distinguishing bots from human game players.

Intel · Software Engineering Intern

2019

Developed Python automation for integration, unit and system-level testing.

GrowMagick · Software Engineer

2019–2020

Built data pipelines and anomaly-detection workflows for an advertising-optimization R&D project.